Online Retail

How is the business performing vs last year?

UK online gift retailer · data 1 Dec 2009 – 9 Dec 2011 · £ GBP

Overview | Customers | Retention | Products | Methodology

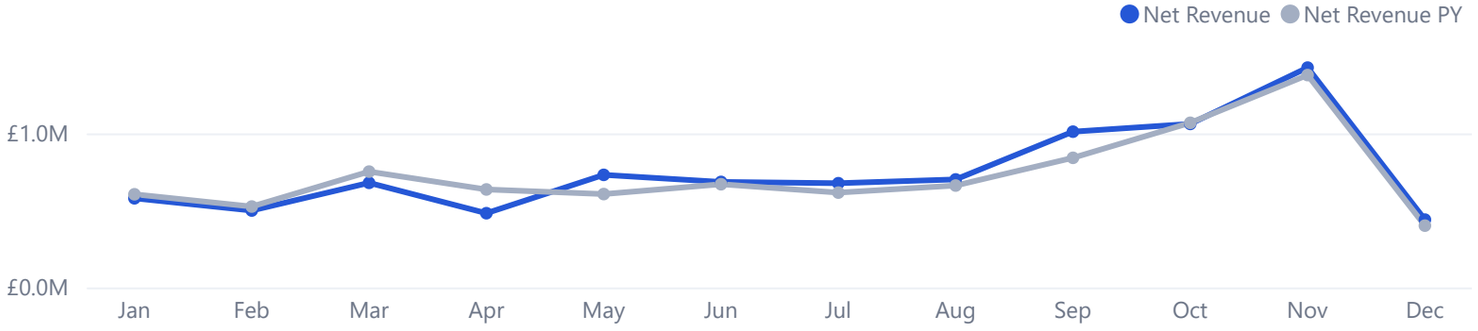
Year: 2011

Market: All

NET REVENUE	ORDERS	ACTIVE CUSTOMERS	AVG ORDER VALUE	RETURN RATE
£8.97M	18,220	4,213	£504.21	2.3%
▲ 2.4% vs same period LY	▼ 3.6% vs same period LY	▲ 1.0% vs same period LY	▲ 6.0% vs same period LY	▼ 0.22 pts vs same period LY

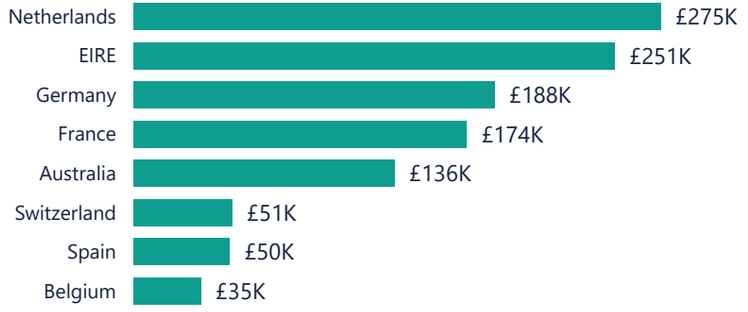
Net revenue by month · 2011 vs 2010

Same calendar month one year earlier · Dec 2011 covers 1–9 Dec only



International markets · UK = 84% of net revenue

Top 8 countries outside the UK, net revenue



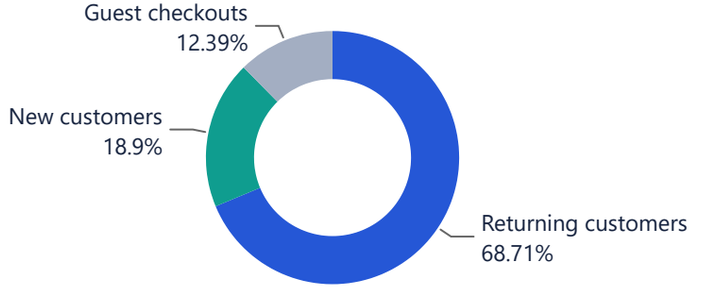
When customers order · orders by weekday and hour

Darker = more orders · the shop is effectively closed on Saturdays

Weekday	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Mon		4	85	257	319	321	462	413	361	319	215	85	2		
Tue		5	117	296	386	399	527	454	391	394	196	110	8		
Wed		5	107	272	405	392	569	461	416	385	226	94		2	1
Thu	1	7	110	276	414	351	544	448	419	399	314	234	162	121	15
Fri		6	120	262	425	364	448	401	372	268	142	78	1	9	1
Sun				2	212	360	425	337	290	316	106				

Where revenue comes from

Gross sales by customer type



Online Retail

Who are our most valuable customers?

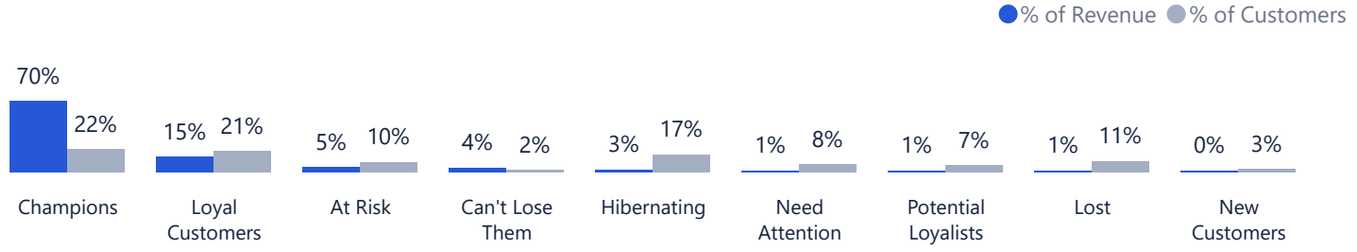
UK online gift retailer · data 1 Dec 2009 – 9 Dec 2011 · £ GBP

Customer market: All

IDENTIFIED CUSTOMERS	REPEAT CUSTOMER RATE	REVENUE PER CUSTOMER	REVENUE FROM TOP 20%	REVENUE FROM CHAMPIONS
5,852	72.3%	£2,786	76.7%	69.8%

RFM segments · share of revenue vs share of customers

All-time, identified customers · RFM snapshot 10 Dec 2011



RFM grid · customers by recency (R) and frequency (F)

5 = most recent / most frequent · top-right = best

R	1	2	3	4	5
5	52	109	184	260	566
4	114	170	247	304	335
3	196	189	269	318	198
2	298	316	285	215	56
1	511	386	185	73	16

Segment playbook · who they are and what to do next (sorted by share of revenue)

Segment	Recommended Action	Customers	% of Customers	Customer Net Revenue	% of Revenue	Avg Recency (days)	Avg Orders	Avg Spend
Champions	VIP rewards, early access to new ranges, reviews	1,303	22.3%	£11,381,799	69.8%	20	16.8	£8,735
Loyal Customers	Upsell higher-value products, loyalty programme	1,202	20.5%	£2,477,167	15.2%	73	5.7	£2,061
At Risk	Personalised win-back campaign	575	9.8%	£800,843	4.9%	368	4.0	£1,393
Can't Lose Them	Direct outreach, account manager call	110	1.9%	£685,450	4.2%	338	12.8	£6,231
Hibernating	Low-cost reactivation, otherwise deprioritise	1,023	17.5%	£428,257	2.6%	381	1.6	£419
Need Attention	Time-limited offers based on past purchases	446	7.6%	£179,014	1.1%	109	1.6	£401
Potential Loyalists	Offer membership, recommend related products	395	6.7%	£175,816	1.1%	27	2.1	£445
Lost	Exclude from paid campaigns	633	10.8%	£127,981	0.8%	565	1.1	£202
New Customers	Onboarding sequence, second-order incentive	165	2.8%	£49,484	0.3%	30	1.0	£300

Online Retail

Do new customers come back?

UK online gift retailer · data 1 Dec 2009 – 9 Dec 2011 · £ GBP

Customer market: All

MONTH-1 RETENTION	MONTH-3 RETENTION	MONTH-6 RETENTION	MONTH-12 RETENTION
21.3%	22.0%	18.5%	19.6%

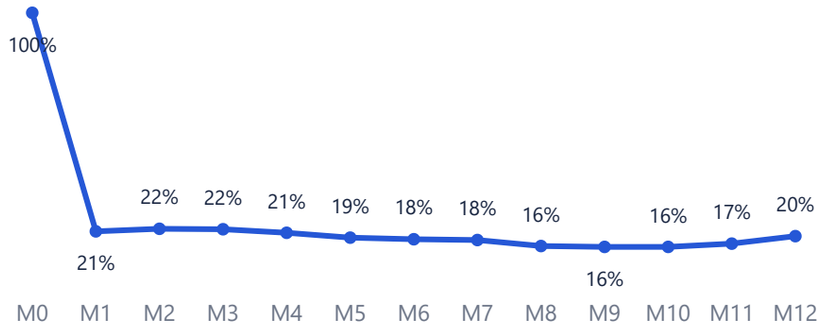
Cohort retention · % of each monthly cohort that ordered again

Rows = month of first purchase (cohort size) · columns = months later · complete months only

Cohort	M0	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Dec 2009 (n=951)	100%	35%	33%	42%	38%	36%	38%	34%	34%	36%	42%	50%	38%
Jan 2010 (n=368)	100%	21%	32%	32%	27%	31%	27%	23%	29%	33%	31%	18%	23%
Feb 2010 (n=375)	100%	23%	23%	29%	25%	20%	19%	29%	26%	28%	11%	13%	15%
Mar 2010 (n=441)	100%	19%	23%	24%	23%	20%	25%	31%	28%	11%	11%	15%	20%
Apr 2010 (n=294)	100%	19%	19%	16%	18%	22%	28%	27%	11%	11%	7%	14%	14%
May 2010 (n=255)	100%	16%	17%	18%	18%	25%	21%	13%	6%	8%	11%	13%	15%
Jun 2010 (n=267)	100%	18%	19%	21%	23%	28%	13%	9%	8%	11%	10%	14%	15%
Jul 2010 (n=185)	100%	16%	18%	30%	29%	14%	11%	15%	15%	11%	14%	15%	13%
Aug 2010 (n=163)	100%	20%	29%	33%	17%	12%	10%	13%	13%	13%	13%	12%	15%
Sep 2010 (n=239)	100%	23%	23%	13%	9%	10%	14%	10%	13%	13%	12%	10%	23%
Oct 2010 (n=375)	100%	26%	15%	13%	9%	8%	13%	14%	11%	9%	11%	13%	19%
Nov 2010 (n=326)	100%	17%	10%	10%	8%	9%	13%	10%	9%	9%	11%	15%	25%
Dec 2010 (n=76)	100%	9%	5%	9%	12%	7%	5%	9%	5%	3%	9%	20%	
Jan 2011 (n=72)	100%	17%	22%	19%	22%	15%	15%	13%	11%	21%	25%		
Feb 2011 (n=125)	100%	16%	16%	18%	22%	16%	16%	15%	15%	18%			
Mar 2011 (n=179)	100%	18%	22%	20%	22%	15%	21%	20%	24%				
Apr 2011 (n=106)	100%	25%	20%	20%	18%	24%	18%	25%					
May 2011 (n=111)	100%	23%	24%	16%	22%	21%	26%						
Jun 2011 (n=108)	100%	23%	21%	27%	20%	28%							
Jul 2011 (n=101)	100%	21%	30%	28%	35%								
Aug 2011 (n=107)	100%	26%	30%	26%									
Sep 2011 (n=188)	100%	27%	38%										
Oct 2011 (n=221)	100%	32%											
Nov 2011 (n=191)	100%												

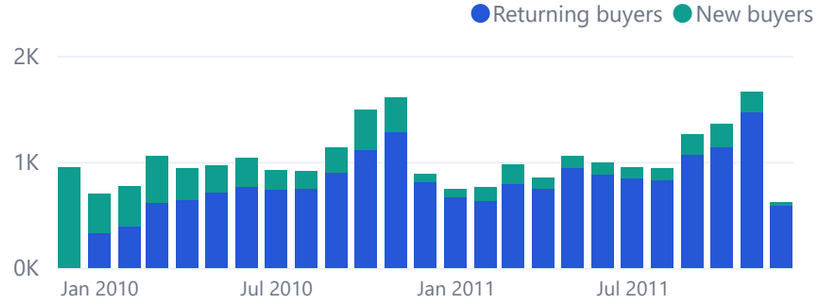
Average retention curve

Average across cohorts with complete data for that month



Active customers per month · returning vs new

Identified customers · Dec 2011 is a partial month



SOURCE: UCI MACHINE LEARNING REPOSITORY – ONLINE RETAIL II (CHEN, 2019) · CLEANED WITH PYTHON, MODELLED IN POWER BI

DASHBOARD BY TOMER ELMAKYES

Online Retail

Which products drive revenue and returns?

UK online gift retailer · data 1 Dec 2009 – 9 Dec 2011 · £ GBP

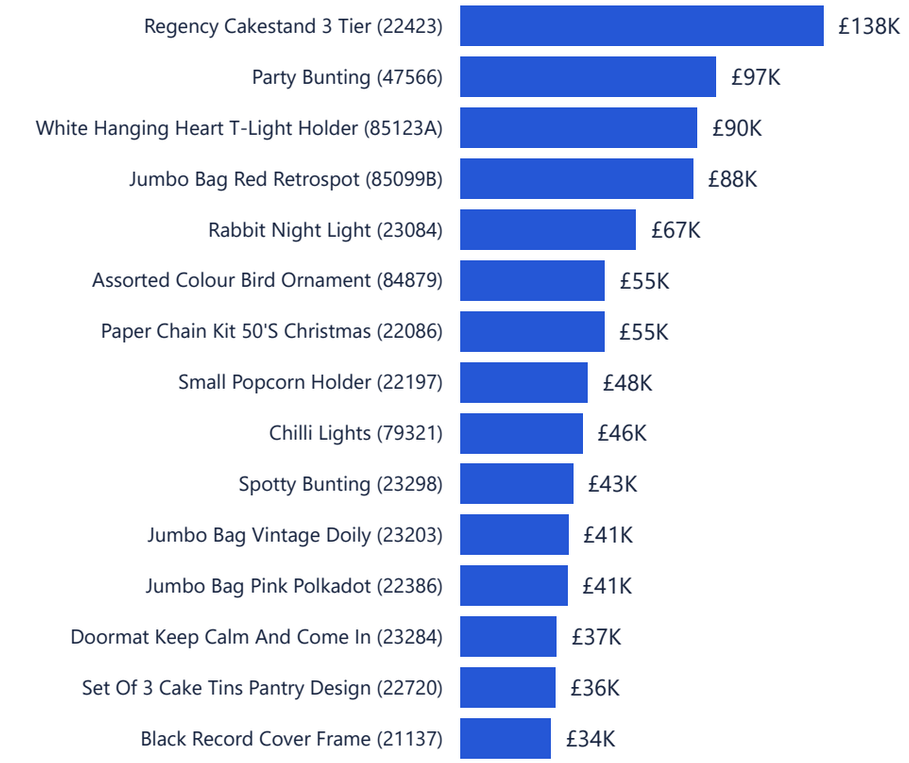
Year: 2011

Market: All

PRODUCTS SOLD	UNITS SOLD	AVG SELLING PRICE	RETURN RATE	PRODUCTS = 80% OF REVENUE
3,710	5.05M	£1.82	2.3% ▼ 0.22 pts vs same period LY	1,039 (22%)

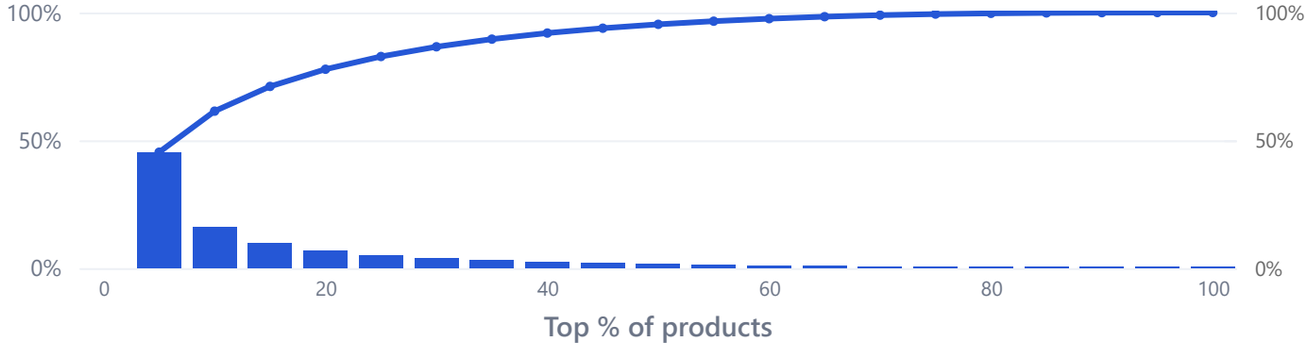
Top 15 products by net revenue

Selected year and market



Pareto · revenue concentration across products (all-time)

Products ranked by net revenue in 5% bands · bars = band share, line = cumulative share · top 20% of products = 78% of revenue



Products with the highest return rate

Products with at least £5K gross sales in the selection

Product	Gross Sales	Returns Value	Return Rate (min £5K sales)
Pantry Chopping Board (23113)	£5,869	£4,803	81.8%
Fairy Cake Flannel Assorted Colour (21108)	£17,386	£6,591	37.9%
Feltcraft Doll Molly (22273)	£11,549	£3,504	30.3%
Tea Time Party Bunting (47566B)	£12,563	£3,693	29.4%
Vintage Red Kitchen Cabinet (22655)	£6,285	£1,436	22.8%
Ivory Diner Wall Clock (22191)	£11,597	£2,603	22.4%

Online Retail

How the data was cleaned and modelled

UK online gift retailer · data 1 Dec 2009 – 9 Dec 2011 · £ GBP

About the data

1.07M invoice lines from a UK online gift retailer (mostly wholesale buyers), 1 Dec 2009 – 9 Dec 2011.
5,852 identified customers in 41 countries and 4,720 products. Guest checkouts have no customer ID.

Metric definitions

Net revenue = gross sales – cancellations (the headline revenue figure)
Orders = distinct invoices Avg order value = gross sales ÷ orders
Return rate = value of cancelled lines ÷ gross sales
vs same period LY = identical dates one year earlier, capped at the last sale (9 Dec)
Active customers = identified customers with at least one order in the period
RFM = Recency, Frequency, Monetary quintile scores (1–5) as of 10 Dec 2011
Cohort = month of a customer's first purchase; retention = share ordering again in month k
ABC class = A: products making up 80% of all-time net revenue, B: next 15%, C: last 5%

RFM segment rules (applied in order)

Champions R ≥ 4 and avg(F, M) ≥ 4
Can't Lose Them R ≤ 2 and avg(F, M) ≥ 4.5
At Risk R ≤ 2 and avg(F, M) ≥ 3
Loyal Customers R ≥ 3 and avg(F, M) ≥ 3
New Customers R ≥ 4 and F = 1
Potential Loyalists other R ≥ 4
Need Attention R = 3
Lost R = 1 and avg(F, M) ≤ 1.5
Hibernating everyone else

Caveats

December 2011 contains 9 days of data; it is excluded from cohort retention (complete months only).
Guest checkouts (14% of 2011 net revenue) are included in revenue but not in customer metrics.
Year-over-year uses the same dates (1 Jan – 9 Dec) so that 2011 is not compared with a full 2010.

Key findings

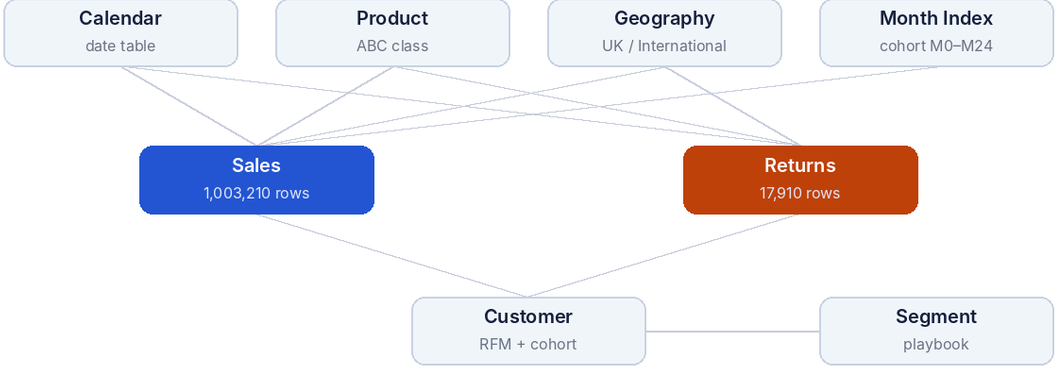
- 2011 net revenue £8.97M, +2.4% vs the same period of 2010: order value rose 6.0% while orders fell 3.6%.
- Champions (22% of customers) generate 70% of revenue; the top 20% of customers generate 77%.
- About 21% of a new cohort orders again the next month, and retention holds at 18–22% up to month 12.
- 1,039 products (22%) make up 80% of revenue · the UK accounts for 84% of 2011 net revenue.
- Orders peak midweek around midday; the shop barely trades on Saturdays.

Data cleaning steps · 1,067,371 raw rows → 1,003,210 clean sales lines

Step	Rule	Rows Removed	Rows After	Note
1	Raw file	0	1,067,371	UCI Online Retail II, Dec 2009 - Dec 2011
2	Remove exact duplicate rows	34,335	1,033,036	Identical invoice lines entered twice
3	Keep real products only	5,980	1,027,056	Drops postage, bank charges, Amazon fees, bad-debt and manual adjustments, tests, vouchers
4	Split cancellations into Returns table	17,914	1,009,142	17,914 cancellation lines kept separately to compute net revenue
5	Remove zero/negative quantity or price	5,928	1,003,214	Stock write-offs and free samples without a sale price
6	Remove reversed bulk orders	4	1,003,210	4 orders over 10K GBP fully cancelled within 30 days (data-entry errors), removed with their cancellations

Data model · star schema

Two fact tables share conformed dimensions · one-to-many relationships, single filter direction



+ Data Quality Log (cleaning steps) · _Measures table: 63 measures in 7 display folders